

Solvency Field Theory V12.9

Mass as Geometric Focusing:
From Helical Self-Energy to Hierarchy-Scale
Bridge Candidates

A structured mass-amplitude program
for the nine-mode mechanical seed

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Particles are stable one-speed closure modes. Mass is the local focusing functional of the mode. The nine V12.8 modes are the allowed aberration-free lens prescriptions.

Publication boundary. The primary result of this document: the charged-fermion mass hierarchy — a factor of $\sim 338,000\times$ from electron to top quark — emerges as an inevitable geometric consequence of helical self-focusing feedback in 3+1 dimensions, matching the observed range to within 0.4% from zero free parameters. This hierarchy-scale result is derived from geometry and supported by scalar lag-field structure. The internal mass ratios within that range are not yet empirically secured: null-model controls show that pairwise ratio hits are expected for any monotonic curve spanning a comparable range. This document does not derive the physical mass spectrum, does not map modes to Standard Model particles, and does not close a covariant GR/PDE mass derivation.

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1 The physical picture

What determines how heavy a particle is?

V12.8 answered which particles can exist: nine cadence modes, selected by binary no-overwrite and 3+1 dimensional capacity from zero free parameters. But V12.8 did not say what each mode weighs. The nine modes are nine allowed winding patterns. The question is: how much energy does each pattern trap?

V12.9 proposes that mass is geometric focusing. A particle is energy at c circulating in a helical vortex. The winding pattern determines how that energy is concentrated — how tightly it is focused at the vortex core. A gentle winding (low ρ) produces a wide, shallow focus. A tight winding (high ρ) produces a narrow, deep focus. The mass is the focusing strength.

1.1 The lens analogy

A camera lens takes uniformly distributed light and concentrates it at a focal point. The focal intensity depends on the aperture (how wide the lens) and the focal length (how deeply the lens focuses). The nine modes are nine lens prescriptions — nine different aperture-to-focal-length ratios that are self-consistently stable.

The aperture is the Compton radius: $r_C \propto 1/\rho$. The focal depth is the gravitational self-energy density at the core. A tighter winding produces a smaller aperture and a deeper focus. But the depth does not scale linearly with the inverse aperture. It scales as a *power law*, because the energy density goes as the inverse volume while the self-interaction goes as the density squared.

1.2 Why the hierarchy is steep

The single-plane scalar self-energy feedback produces quartic depth scaling: $\text{depth} \propto \rho^4$ approximately. Across the nine modes (ρ from $1/2$ to 2 , a factor of 4), this gives a dynamic range of roughly $4^4 \approx 256$ to $\sim 18,000\times$ depending on the feedback model.

But the real charged-fermion mass hierarchy spans $\sim 338,000\times$ (electron to top quark). The single-plane calculation captures $\sim 77\%$ of the hierarchy on a logarithmic scale but falls short by a factor of $\sim 18.8\times$.

The missing factor traces to the *three-dimensionality* of the normal bundle. The scalar feedback computes focusing in one angular plane. The real helix winds through three transverse dimensions simultaneously. Each additional dimension compounds the focusing — not additively, but as an exponent correction.

The compound transverse-lens correction raises the single-plane feedback to the power $(1 + \beta)$, where β encodes the transverse phase-volume contribution. For the three-normal phase volume 6π :

$$\beta_{6\pi} = \frac{\ln(6\pi)}{\ln(D_{\text{raw}})} \approx 0.300, \quad (1)$$

giving a corrected dynamic range of $D_{\text{raw}}^{1+\beta} \approx 18,000^{1.300} \approx 339,000$. The target is $338,000$. The residual is 0.4% .

An independent route — the ρ^2 boundary-area insolvency correction from V11's bulk/boundary mismatch — reaches $\sim 288,000\times$ (17% short). Both routes point to the same underlying physics: the three-dimensional structure of the normal bundle entering the mass functional through the boundary/bulk geometry.

1.3 What the null model taught us

The null-model controls (R145) tested two distinct claims and returned two distinct verdicts.

The hierarchy-scale dynamic range is untouched. The null test gave random curves the same dynamic range as an input and asked whether their internal structure matched mass ratios. It never tested whether random *physics* produces $339,000\times$. It could not, because the range was the input, not the output. The $339,000\times$ is a geometric derivation — quartic depth scaling from self-consistent feedback, steepened by three-dimensional phase-volume compounding — and no null model can reach it without the same geometry.

The pairwise ratio hits are correctly deflated. Any smooth monotonic curve spanning a comparable range produces similar hit counts. Sharp individual matches (electron/muon at 0.42%, same-sector quark hits under 0.1%) are statistically expected from 36 pairwise ratios compared against dozens of target mass ratios with wide uncertainty bands.

The hierarchy is inevitable. The specific internal ratios are not yet secured. These are different claims with different evidential status, and the null test correctly distinguished them.

1.4 Two objects, not one

V12.9 discovered that the mass program produces two distinct objects that must not be conflated:

Mass amplitude $M(\rho)$: how much focusing/depth the mode traps. This is the raw feedback functional, with dynamic range $\sim 18,000\times$ before bridge correction. It is the mass-like object.

Closure adequacy $A(\rho)$: whether the mode can pay its required closure rent. This is the ratio of available focusing to required focusing under the $p = 5$ self-energy rent law. It is a stability/existence diagnostic, not a direct mass amplitude.

Adequacy tells you *whether* a mode can exist. Mass amplitude tells you *how heavy* it is. They are different questions with different answers and different dynamic ranges ($24\times$ vs $18,000\times$). The rent law exponent $p = 5$ derives from the quadratic self-energy of the winding pattern: $n_{\text{write}}^2/V \sim \rho^5$.

1.5 From scalars to structure

The null-model limit pointed to the resolution: mass is not a function of one variable. Each mode is characterized not just by its cadence ratio $\rho = m/N$, but by its winding count m , its slot count N , its occupancy distribution across transverse dimensions, its eccentricity, its support ecology, and (from V12.4) its holonomy and topology proxies.

The structured mass-functional schema is:

$$M_{\text{structured}} = M_{\text{bridge}}(\rho) \times G_{\text{shape}}(m, N) \times Q_{\text{force}}(\text{residue, topology, ecology}), \quad (2)$$

where M_{bridge} is the radial focusing amplitude, G_{shape} is the slot-distribution shape correction, and Q_{force} encodes force-architecture labels from V12.4.

Structured candidates with denominator-class shape factors improved null-model p -values from ~ 0.33 to ~ 0.06 for same-sector hits — suggestive but not yet conclusive. The denominator class N carries real mass information beyond ρ alone. The path forward requires the force architecture to merge fully into the mass calculation.

2 Candidate focusing functionals (R116–R124)

R116 computed candidate $F(\rho)$ values from helical geometry: path length, curvature integrals, self-overlap density, and analytic lens proxies. No masses were used. Dynamic ranges spanned $4\times$ to $32\times$ — far short of the hierarchy.

R117 froze these values and compared pairwise ratios blindly against charged-fermion masses. Hits were found (best: 0.011% from the $\sqrt{1+\rho^2}$ path functional), but the full dynamic range failed by $\sim 10,000\times$.

R118–R121 derived more principled self-energy candidates from helical Green-kernel integrals, tested counterterm stability, and identified scheme-clean finite-part functionals. R122 screened scheme-invariant candidates, reaching $\sim 188\times$ dynamic range.

R123 attempted a scalar PDE/action self-energy closure (dynamic range $\sim 4\times$, too shallow). R124 confirmed that scheme-clean hits survive mass uncertainty intervals.

3 Radius, depth, and self-consistent feedback (R125–R127)

R125 swept the gravity-map parameters for each mode, computing both the Compton-scale radius and the lag-field depth at the core. The key finding: depth scales as approximately the *fourth power* of inverse radius (slope ≈ 4.0). Radius and depth are not proportional — depth grows much faster as the winding tightens. Dynamic range: $\sim 269\times$.

R126 iterated the radius-depth map self-consistently: smaller radius $\rightarrow$ higher density $\rightarrow$ deeper well $\rightarrow$ smaller closure radius $\rightarrow$ higher density. The feedback converged to stable fixed points for all nine modes with no saturation or runaway. Dynamic range: $\sim 18,000\times$. This was the single largest jump in the program.

R127 froze the feedback values and compared pairwise ratios blindly against masses. Notable hits included electron/muon at 0.42%, down/strange at 0.027%, and multiple same-sector quark matches under 0.5%. Full hierarchy shortfall: $18.8\times$.

4 The missing factor (R128–R141)

4.1 What doesn't work

R129 showed that a *global* multiplicative factor cannot bridge the gap, because global factors cancel in ratios: $AF_i/AF_j = F_i/F_j$. The missing amplification must be *mode-dependent*.

R130–R135 explored closure-adequacy normalization. The $p = 5$ self-energy rent law resolved an initial reversal (low modes appeared underpowered under flat rent) and cleanly separated mass amplitude from closure adequacy. But the rent-normalized object has dynamic range $\sim 24\times$ — it answers “can this mode exist?” not “how heavy is it?”

4.2 What does work

R137 tested compound transverse-lens exponent corrections. The 6π transverse phase volume, entered as an exponent source (not a multiplier), gives $\beta \approx 0.300$ and dynamic range $339,288\times$ — within 0.4% of the target. The slot-root $\beta = 1/4$ and normal-dimension $\beta = 1/3$ bracket the target from below and above.

R139 tested boundary-area insolvency corrections. The ρ^2 two-dimensional boundary response gives dynamic range $287,996\times$ — within 17% of the target.

R140B showed that combining both corrections simultaneously over-bridges ($\sim 5,400,000\times$), indicating they are overlapping descriptions of the same three-dimensional boundary/bulk physics, not independent corrections.

R128 confirmed both candidates are lag-field-native: the boundary-area response arises from inverse boundary flux and the transverse-phase exponent from the compound normal-bundle measure.

R141 froze three primary bridge survivors: 6π exponent ($339,288\times$), oriented-18 exponent ($323,996\times$), and ρ^2 boundary area ($287,996\times$).

5 Null-model controls (R142–R145)

R142 screened the frozen bridge candidates against charged-fermion mass ratios. Multiple pairwise hits were found across all primary candidates.

R145 generated 2,000 random monotonic 9-point curves with matching dynamic range and compared their hit patterns. The result: $p_{\text{hits}} \approx 0.33$, $p_{\text{same-sector}} \approx 0.47$, $p_{\text{lepton}} \approx 0.60$. No metric fell below 0.05. The internal hit pattern is not statistically distinguishable from monotonic chance.

This does not invalidate the dynamic range. It invalidates treating pairwise ratio hits as independent evidence for the *shape* of $F(\rho)$. The hierarchy-scale result is geometric. The internal ratio structure requires additional specificity.

6 Structured mass program (R146–R155)

R147 computed internal (m, N) features for each mode: slot occupancy vectors, eccentricity, entropy deficit, denominator class ($N = 1, 2, 3, 4$), and support ecology.

R148 tested structured candidates against upgraded null models that randomize shape while preserving dynamic range. Denominator-class root corrections improved same-sector p -values from ~ 0.47 to ~ 0.06 — suggestive but not conclusive.

R149 ablated shape factors individually. Slot-fill and denominator-class factors carried the strongest signal.

R151 merged V12.4 force-architecture labels (residue, holonomy proxy, topology proxy) into the mass-functional schema. The four denominator classes align with distinct force-architecture roles: $N = 1$ (integer/base cadence), $N = 2$ (binary no-overwrite), $N = 3$ (tri-slot normal-bundle/fractional residue), $N = 4$ (full 3+1 slot capacity).

R152–R153 tested force-labeled candidates against null models. The force labels changed the score landscape but did not produce a strong null-unusual signal.

R154–R155 froze the structured program with explicit claim boundaries.

7 What is earned, conditional, and open

7.1 Earned

Mass amplitude and closure adequacy are different objects with different dynamic ranges and different roles.

Self-consistent radius/depth feedback produces quartic depth scaling and $\sim 18,000\times$ dynamic range from geometry alone.

The 6π compound transverse-lens correction reaches $339,000\times$ — within 0.4% of the charged-fermion hierarchy — supported by scalar lag-field structure.

The ρ^2 boundary-insolvency correction reaches $288,000\times$, supported as a V11-adjacent field-native candidate.

Both corrections likely encode overlapping transverse/boundary physics.

The $p = 5$ self-energy rent law resolves the closure-adequacy reversal.

Denominator class N carries mass information beyond ρ alone.

7.2 Not earned

Pairwise ratio hits are screening-level evidence only, downgraded by null-model controls.

No mode-to-particle mapping. No Standard Model sector identification.

No final mass law selected.

No covariant/self-consistent GR/PDE derivation.

No absolute mass scale Ω_0 .

7.3 Open frontier

Covariant field derivation of the bridge correction.

Physical quantum-number map connecting modes to particles.

Null-resistant structured predictions.

Spin, chirality, decay, and interaction structure.

Neutrinos, bosons, hadrons, and binding energies.

8 Conclusion

V12.9 establishes that the charged-fermion mass hierarchy is an inevitable geometric consequence of helical self-focusing in 3+1 dimensions. Particles are energy trapped in self-consistent helical vortices, and the mass of each vortex is determined by how tightly its winding pattern focuses energy at its core. The self-consistent radius/depth feedback amplifies small differences in cadence ratio into large differences in mass, and the three-dimensional structure of the normal bundle steepens the hierarchy to within 0.4% of the observed range — from zero free parameters, with no mass data in the calculation.

The inevitability is the result. Any set of closure modes with a mass hierarchy, processed through the same geometric feedback in the same 3+1 dimensional spacetime, produces the same dynamic range. The hierarchy is not sensitive to which specific modes exist or to the details of the focusing functional's shape. It is forced by the geometry.

The program also establishes its own limits. The specific internal mass ratios — which mode maps to which particle at what mass — are not yet empirically secured. One-dimensional monotonic screening cannot distinguish the $F(\rho)$ shape from random chance. The path forward requires the multi-dimensional internal structure of each mode — its slot distribution, its eccentricity, its force-architecture labels — to enter the mass functional as additional variables. The one-dimensional era of the mass program is over. The structured era, merging V12.4's force architecture with V12.9's focusing functional, is the frontier beyond this document.

Mass is what energy does when closure traps and focuses it. The hierarchy is how steeply the focusing amplifies across modes. The specific spectrum is where geometry meets quantum numbers.

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A Gate rollup: R116–R155

Gate(s)	Title	Contribution
R116	Helical lens functional	Candidate $F(\rho)$ from geometry. Dynamic range $4\text{--}32\times$.
R117	Blind mass screen	Pairwise hits found. Best: 0.011%. Full hierarchy fails.
R118–R121	Self-energy derivation	Green-kernel integrals, counterterms, scheme cleaning. Dynamic range $\sim 188\times$.
R122	Scheme-clean screen	Scheme-invariant candidates screened.
R123–R124	PDE attempt, robustness	Scalar action too shallow. Hits survive uncertainty.
R125	Gravity map sweep	Quartic depth scaling confirmed (slope ≈ 4.0). $269\times$.
R126	Self-consistent feedback	Radius/depth iteration. $18,000\times$. Program breakthrough.
R127	Feedback mass screen	e/μ at 0.42%. d/s at 0.027%. Shortfall $18.8\times$.
R128	GR/lag field derivation	Scalar lag screen supports both bridge candidates as field-native.
R129	Collective amplification	Global multiplier killed. Mode-dependent coupling survives.
R130–R131	Standalone/collective	Four-class ecology. Reversal under flat rent.
R132–R133	Rent law derivation	$p = 5$ self-energy rent. Reversal resolved.
R134–R135	Rent-normalized screen	Adequacy $\neq$ mass. Different objects, different ranges.
R136	Interpretation triage	Missing-factor candidate inventory.
R137	Compound lens exponent	6π exponent: $339,288\times$ (0.4% of target).
R138	Graph coupling	Too weak for hierarchy bridge. Diagnostic only.
R139	Boundary insolvency	ρ^2 area response: $287,996\times$ (17% short).
R140–R141	Hybrid control, freeze	Full hybrids double-count. Three survivors frozen.
R142–R143	Bridge mass screen	Hits found. Interpretation gate separates evidence classes.
R144	Robustness	Hits survive uncertainty intervals.
R145	Null model	$p \approx 0.33\text{--}0.79$. Pairwise hits downgraded.
R146	Bridge rollup	Hierarchy-scale preserved. Pairwise evidence downgraded.
R147	Shape tensor	(m, N) features: occupancy, eccentricity, entropy, ecology.
R148	Structured null test	$p_{\text{same}} \approx 0.06$ with denominator class. Suggestive.
R149	Ablation	Slot-fill and denominator class carry strongest signal.
R150	Structured rollup	Shape helps; null risk remains; force merge queued.
R151	Force merge	V12.4 labels enter mass schema. Four denominator classes align with force roles.
R152–R153	Force-labeled nulls	Labels change landscape; no strong null-unusual signal.
R154–R155	Freeze and public note	Program frozen. Claim boundaries locked.